Priority vs SRTF Comparison Project

Rules

Part	Used Rule
Priority	Lower number means higher priority. Tie break: arrival time, then input order.
SRTF	Shortest remaining burst time runs first. Preemption happens when a shorter job arrives.
Metrics	Waiting Time, Turnaround Time, Response Time, and their averages.

Test Cases

Scenario A: Basic Mixed Workload

Focus: Different arrival times, burst times, and priorities.

PID	Arrival	Burst	Priority
P1	0	5	3
P2	1	3	1
P3	2	8	4
P4	3	2	2

Priority Gantt Chart

P1	P2	P4	P1	P3
0-1	1-4	4-6	6-10	10-18

SRTF Gantt Chart

P1	P2	P4	P1	P3
0-1	1-4	4-6	6-10	10-18

Algorithm	Avg WT	Avg TAT	Avg RT
Priority	3.50	8.00	2.25
SRTF	3.50	8.00	2.25

Scenario B: Priority vs Burst Time

Focus: A high-priority long process runs against short low-priority processes.

PID	Arrival	Burst	Priority
P_Long_HighPri	0	10	1
P_Short_LowPri	1	2	10
P_Medium	2	4	5

Priority Gantt Chart

Long 0-10	Med 10-14	Short 14-16

SRTF Gantt Chart

Long 0-1	Short 1-3	Med 3-7	Long 7-16

Algorithm	Avg WT	Avg TAT	Avg RT
Priority	7.00	12.33	7.00
SRTF	2.33	7.67	0.33

Scenario C: Starvation-Sensitive Case

Focus: Shows how low-priority short processes can wait behind a long urgent process.

PID	Arrival	Burst	Priority
P1_HighPri_Long	0	15	1
P2_LowPri_Short	1	2	5
P3_LowPri_Short	2	2	5
P4_LowPri_Short	3	2	5
P5_LowPri_Short	4	2	5

Priority Gantt Chart

P1 0-15	P2 15-17	P3 17-19	P4 19-21	P5 21-23

SRTF Gantt Chart

P1 0-1	P2 1-3	P3 3-5	P4 5-7	P5 7-9	P1 9-23

Algorithm	Avg WT	Avg TAT	Avg RT
Priority	12.40	17.00	12.40
SRTF	2.80	7.40	1.20

Scenario D: Validation Case

Invalid Input	Expected Handling
PID: P_Error, Arrival: 0, Burst: -5, Priority: 99	Show an error before simulation starts.
Empty or duplicate PID	Show an error and reject the process.
Negative arrival time	Reject the process data.
Zero or negative burst time	Reject the process data.
Invalid priority value	Show an error for the priority field.

Analysis

Question	Answer
Lower average waiting time?	SRTF gives the lower average waiting time in the conflict cases.
Lower average response time?	SRTF is lower in Scenario B, while Priority is lower for urgent long jobs.
Did priority help urgent processes?	Yes. High-priority long processes start quickly under Priority.
Did SRTF favor short jobs?	Yes. Short jobs finish earlier when they arrive.
Recommended algorithm?	Use SRTF for efficiency. Use Priority when urgent processes matter most.

Conclusion

Point	Conclusion
Best overall metrics	SRTF gives better averages in the conflict cases.
Priority advantage	Priority gives better service to urgent processes.
Main trade-off	Efficiency for short jobs vs policy-based service.
Fairness	SRTF is fairer to short jobs. Priority is fairer to urgent jobs.